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(Cyber Security)

Wireless Security

(CYS331)

Unit 7

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Wireless Security Standards

- Wireless security standards are protocols designed to protect wireless networks (Wi-Fi) from unauthorized access, data breaches, and interference. These standards evolve over time to address emerging threats and improve encryption mechanisms.

- 1. WEP (Wired Equivalent Privacy)
- 2. WPA (Wi-Fi Protected Access)
- 3. WPA2 (Wi-Fi Protected Access II)
- 4. WPA3 (Wi-Fi Protected Access III)
- 5. IEEE 802.11i
- 6. IEEE 802.1X

1. WEP (Wired Equivalent Privacy)

Background:

- Introduced in 1997 as part of the original Institute of Electrical and Electronics Engineers (IEEE) 802.11 standard
- Goal: Provide privacy similar to that of wired networks

Technical Details:

- **RC4**: Rivest Cipher 4, a stream cipher used for encryption
- **IV**: Initialization Vector, 24-bit field added to the encryption key
- Key length:
  - 64-bit (40-bit key + 24-bit IV)
  - 128-bit (104-bit key + 24-bit IV)
- Authentication methods:
  - **Open System**: Any device can authenticate
  - **Shared Key**: Devices authenticate using a shared secret

WEP (Wired Equivalent Privacy) Cont

Weaknesses:

- IVs are too short and frequently reused
- RC4 is vulnerable to statistical attacks
- Key reuse makes WEP susceptible to replay attacks

Real-World Attack:

- Tools like Aircrack-ng can capture encrypted packets and use them to recover WEP keys

Outcome:

- Declared insecure by the Wi-Fi Alliance in 2004

No longer used in secure networks

2. WPA – Wi-Fi Protected Access

- **Background:**
- Introduced in 2003 by the Wi-Fi Alliance
- Interim solution to replace WEP until IEEE 802.11i was finalized
- **Technical Details:**
- **TKIP**: Temporal Key Integrity Protocol
  - Dynamically generates new keys for every packet
  - Includes a MIC (Message Integrity Check) to detect tampering
- Still uses RC4 for encryption
- Supports two modes:
  - WPA-PSK: Pre-Shared Key (for home users)
  - WPA-Enterprise: Uses IEEE 802.1X authentication and a RADIUS (Remote Authentication Dial-In User Service) server

WPA – Wi-Fi Protected Access Cont

- **Weaknesses**:
- TKIP still relies on RC4, inheriting many of its weaknesses
- Vulnerable to replay attacks and MIC key recovery
- **Real-World Attack**:
- **Beck-Tews attack** allows packet injection by exploiting TKIP flaws

3. WPA2 – Wi-Fi Protected Access II

- **Background:**
- Released in 2004, based on the IEEE 802.11i amendment
- Made mandatory for Wi-Fi certified devices by 2006
- **Technical Details:**
- **AES**: Advanced Encryption Standard (used instead of RC4)
- **CCMP**: Counter Mode with Cipher Block Chaining Message Authentication Code Protocol
  - Provides both encryption and message integrity
- Uses a **4-way handshake** to establish encryption keys

Authentication Modes:

- WPA2-PSK:
- WPA2-Enterprise

WPA2 – Wi-Fi Protected Access II Cont

Vulnerability:

- KRACK (Key Reinstallation Attack, 2017): Exploits flaws in the 4-way handshake process to reinstall already-used keys

4. WPA3 – Wi-Fi Protected Access III

Background:

- Introduced in 2018 to overcome WPA2 limitations

Technical Details:

- **SAE**: Simultaneous Authentication of Equals
  - Replaces WPA2's PSK
  - Based on **Password-Authenticated Key Exchange (PAKE)**, providing resistance to dictionary attacks
- **Forward Secrecy**: Past session keys remain secure even if a current key is compromised

WPA3 – Wi-Fi Protected Access III

- **Additional Features:**
- **OWE**: Opportunistic Wireless Encryption
  - Provides encrypted communication in open networks
- **Wi-Fi Easy Connect**:
  - Uses **DPP** (Device Provisioning Protocol) to connect IoT (Internet of Things) devices securely via QR codes
- **Vulnerability:**
- **Dragonblood attacks** (2019): Exploited implementation flaws in the SAE protocol, now mitigated in updates

5. IEEE 802.11i – Wireless LAN Security Amendment

Background:

- Finalized in 2004
- Introduced the Robust Security Network (RSN) concept

Features:

- Introduced AES-CCMP as the default encryption mechanism
- Mandated 802.1X for authentication in enterprise mode
- Defined protocols for key management, authentication, and data integrity

6. IEEE 802.1X – Port-Based Network Access Control

Role

- Provides a framework for authenticating and authorizing devices on a network

Components:

- **Supplicant**: The client device requesting access
- **Authenticator**: Network device (e.g., Access Point or Switch)
- **Authentication Server**: Usually a RADIUS server

EAP Variants Used:

- **EAP-TLS**: Uses client and server certificates
- **PEAP**: Protected EAP, encapsulates EAP inside a TLS tunnel
- **EAP-TTLS**: Tunneled Transport Layer Security

4-way handshake

- A 4-way handshake is a process used in wireless security protocols, particularly WPA/WPA2/WPA3, to establish a secure connection between a Wi-Fi client (like your laptop or phone) and a Wi-Fi access point (router). It ensures both sides have the correct credentials and creates encryption keys for secure communication.

Purpose of the 4-Way Handshake

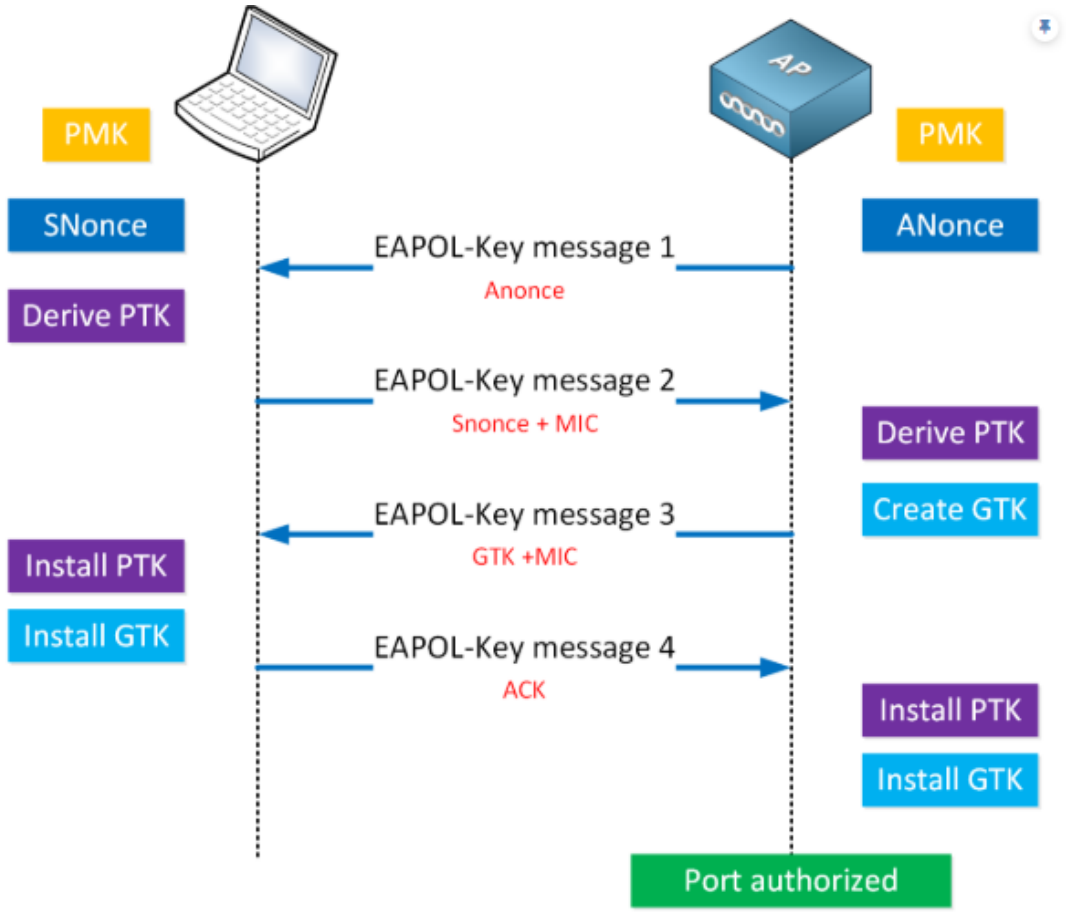
- To confirm both the client and access point know the Pre-Shared Key (PSK) or use 802.1X authentication.
- To generate and exchange encryption keys (called Pairwise Transient Keys or PTKs).
- To protect against replay and man-in-the-middle attacks.

4-way handshake cont

- **Steps in the 4-Way Handshake**

1. **Message 1: AP → Client**
    1. The Access Point (AP) sends a **nonce (ANonce)** — a random number — to the client.
    2. This begins the key exchange process.
2. **Message 2: Client → AP**
    1. The client uses the ANonce, its own nonce (SNonce), and both MAC addresses to generate a Pairwise Transient Key (PTK).
    2. It sends the SNonce back to the AP, along with a Message Integrity Code (MIC) to prove it knows the key.
3. **Message 3: AP → Client**
    1. The AP uses the received SNonce to generate the same PTK.
    2. It sends the **Group Temporal Key (GTK)** encrypted using the PTK and includes a MIC to validate.
4. **Message 4: Client → AP**
    1. The client sends an acknowledgment to confirm it successfully received and installed the keys.
    2. Secure communication can now begin.

4-Way Handshake in a Wireless Network



Acronyms

Acronym	Full Form
AES	Advanced Encryption Standard
CCMP	Counter Mode with Cipher Block Chaining Message Authentication Code Protocol
CRC	Cyclic Redundancy Check
EAP	Extensible Authentication Protocol
GCMP	Galois/Counter Mode Protocol
IEEE	Institute of Electrical and Electronics Engineers
IoT	Internet of Things
IV	Initialization Vector
MIC	Message Integrity Check
OWE	Opportunistic Wireless Encryption
PAKE	Password Authenticated Key Exchange
PEAP	Protected Extensible Authentication Protocol
PMF	Protected Management Frames
PSK	Pre-Shared Key
RADIUS	Remote Authentication Dial-In User Service
RC4	Rivest Cipher 4
RSN	Robust Security Network
SAE	Simultaneous Authentication of Equals
SSID	Service Set Identifier
TKIP	Temporal Key Integrity Protocol
TLS	Transport Layer Security
WPA	Wi-Fi Protected Access
WEP	Wired Equivalent Privacy